

Shi Yin

Curriculum Vitae

Institut für Theoretische Physik
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Education and Research Experience

Justus Liebig University Giessen
Humboldt Research Fellow

April 2024 – present
Giessen, Germany

- Host: Dr. Fabian Rennecke

Dalian University of Technology
Ph.D, Theoretical Physics

September 2018 – Jun 2023
Dalian, Liaoning Province, China

- Advisor: Prof. Wei-jie Fu
- Thesis Title: Theoretical study of QCD at finite temperature and density within the functional renormalization group approach

Research Interests

Quantum chromodynamics (QCD) phase structure

QCD at finite temperature and density, Chiral phase transition, Critical phenomena

Relativistic heavy ion collision phenomenology

Baryon number fluctuation, Signatures of critical end point

QCD inhomogeneous phase

Moat regime, Inhomogeneous chiral condensate, Spatial modulation phase

Real-time correlation in QCD

spectral function, non-equilibrium effect

High density strong interaction matter

Equation of state, First principle high density QCD computation

Academic Cooperation

Member of fQCD collaboration: <https://fqcd-collaboration.github.io>

Member of CRC-TR 211: <https://crc-tr211.org>

Service to the Scientific Community

Referee for Phys. Rev. D

Teaching experience

Substitute lecturing for Advanced Quantum Mechanics and Quantum Field Theory

Selected Publications

Full list of publications can be found in INSPIRE:

<https://inspirehep.net/authors/1762169?ui-citation-summary=true>

1. F. Rennecke, S. Yin, (2025) *Dissecting the moat regime at low energies I: Renormalization and the phase structure*, [Phys.Rev.D 113. 074015](#) [[arXiv:2412.15949](#)]
2. W.-j. Fu, J. M. Pawłowski, R. D. Pisarski, F. Rennecke, R. Wen, S. Yin, (2024) *QCD moat regime and its real-time properties*, [Phys.Rev.D 111. 094026](#) [[arXiv:2412.15949](#)]
3. J. Braun, Y.-r. Chen, W.-j. Fu, F. Gao, C. Huang, F. Ihssen, J. M. Pawłowski, F. Rennecke, F. R. Sattler, Y.-y. Tan, R. Wen, S. Yin, (2023) *Soft modes in hot QCD matter*, [Phys.Rev.D 111. 094010](#) [[arXiv:2310.19853](#)]
4. W.-j. Fu, X. Luo, J. M. Pawłowski, F. Rennecke, S. Yin, (2023) *Ripples of the QCD critical point*, [Phys. Rev. D 111. L031502](#) [[arXiv:2308.15508](#)]
5. W.-j. Fu, X. Luo, J. M. Pawłowski, F. Rennecke, S. Yin, R. Wen, (2021) *Hyper-order baryon number fluctuations at finite temperature and density*, [Phys. Rev. D 104. 094047](#) (Editors' Suggestion)[[arXiv:2101.06035](#)]

Invited and Contributed Conference Talks

1. *Dissecting the moat regime at low energies*
CPOD2026 - 16th Workshop on Critical Point and Onset of Deconfinement
CERN, 13 to 17 April 2026
2. *The moat regime at low energies*
CRC retreat 2026 (Highlight talk and roundtable discussion)
Bonn Germany, 23 to 27 March 2026
3. *Dissecting the moat regime at low energies*
DPG conference 2026
Erlangen Germany, 15 to 20 March 2026
4. *Dissecting the moat regime at low energies*
The QCD Phase Diagram: From Theory to Experimental Signatures
Dalian China, 8 to 11 October 2025
5. *Estimates on the convergence of expansions at finite baryon chemical potential*
Analytic structure of QCD and Yang-Lee edge singularity
ETC* Trento Italy, 8 to 12 September 2025 (invited talk)
6. *Moat regimes and baryon fluctuations*
CRC Retreat 2025
Bonn Germany, 17 to 21 March 2025
7. *The moat regime in QCD*
QuantFunc 2024
Valencia Spain, 2 to 6 September 2024

8. *Baryon number fluctuations at large baryon chemical potentials*
CPOD2022 - Workshop on Critical Point and Onset of Deconfinement
28 November 2022 to 2 December 2022
9. *Real time meson two point functions under fRG-QCD*
FunQCD: from first principles to effective theories
13 to 17 June 2022
10. *Scalar and pseudo-scalar meson spectral functions under fRG-QCD*
The 14th Workshop on QCD Phase Transition and Relativistic Heavy-ion Physics (QPT 2021)
Guiyang China, 26 to 30 July 2021 (Third prize of excellent student talk)
11. *Hyper-order baryon number fluctuations at finite temperature and density*
FunQCD: from first principles to effective theories
29 March to 1 April 2021
12. *Hyper-order baryon number fluctuations at finite temperature and density*
10th International Conference on Exact Renormalization Group 2020 (ERG 2020)
Kyoto University, 2 to 6 November 2020
13. *Mesonic dynamics and the QCD phase transition*
The 18th National Conference on Medium and High Energy Nuclear Physics
Changsha China, 22 to 25 June 2019
14. *Nontrivial dispersion relation and QCD thermodynamics in the low energy effective model*
QCD continuous field theory method and its application
Dalian China, 23 to 26 July 2018

Posters

1. *The moat regime in QCD*
The 21st International Conference on QCD in Extreme Conditions (XQCD) 2025
Wroclaw Poland, 2 to 4 July 2025 (poster)
2. *The moat regime in QCD*
Quark Matter 2025
Frankfurt Germany, 6 to 12 April 2025 (poster)

References

1. Prof. Wei-jie Fu

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2. Dr. Fabian Rennecke

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